

Analysis of the Obstacle Potential Function

Abstract

We study the obstacle potential U_{obst} used for collision avoidance, examine the resulting force field $\mathbf{F} = -\nabla U_{obst}$ around the obstacle vehicle, and show that the relative-velocity term $\Delta v / \Delta v_{max}$ causes the force to grow even as the vehicle recedes. We drop this term and re-examine the force response. Finally, we consider an alternative pseudo-distance scaling – combining the per-axis time constants into a single, jointly velocity-scaled term – and compare its behaviour to the original formulation.

1 Potential Function

The obstacle potential is defined as

$$U_{obst} = \frac{\lambda(\Delta v / \Delta v_{max})}{\left(1 - \frac{w}{\Delta x}\right) \sqrt{\left(\frac{\Delta x}{\tau(\Delta x, \Delta v_x)}\right)^2 + \left(\frac{\Delta y}{\tau(\Delta y, \Delta v_y)}\right)^2}} \quad (1)$$

where the direction-dependent time constants are given by

$$\tau(\Delta x, \Delta v_x) = \frac{(1 + \alpha \Delta v_x) + (\alpha \Delta v_x - 1) \tanh(-\Delta x \Delta v_x)}{2} \tau_x \quad (2)$$

$$\tau(\Delta y, \Delta v_y) = \frac{(1 + \alpha \Delta v_y) + (\alpha \Delta v_y - 1) \tanh(-\Delta y \Delta v_y)}{2} \tau_y \quad (3)$$

Here $\Delta x, \Delta y$ are the relative position offsets to the obstacle vehicle, $\Delta v_x, \Delta v_y$ the relative velocities, w the obstacle width, and $\lambda, \tau_x, \tau_y, \alpha, \Delta v_{max}$ are model parameters.

1.1 Force Field

The force field $\mathbf{F} = -\nabla U_{obst}$ resulting from Equation 1 is shown below under three relative-velocity scenarios: a pure lateral closing speed (Δv_x only), a pure longitudinal closing speed (Δv_y only), and both combined.

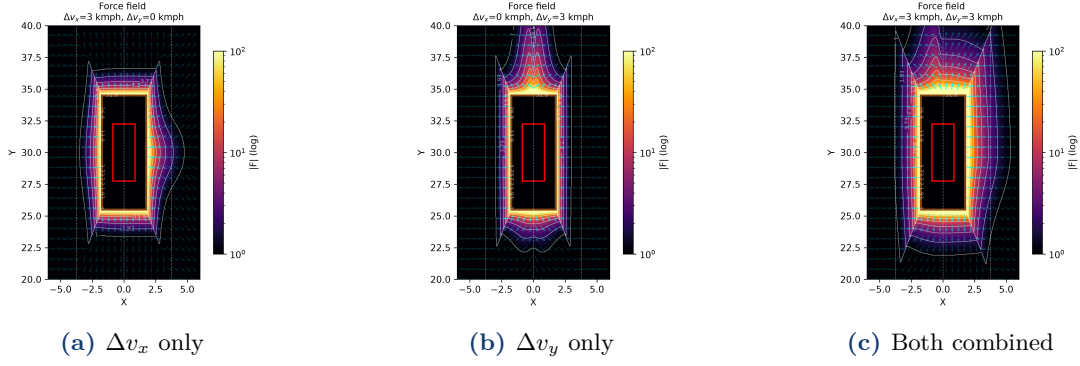


Figure 1: Force field of the potential in Equation 1.

2 Force at a Fixed Point vs. Relative Velocity

To examine the velocity dependence of the force in more detail, we fix the position at $(x, y) = (3, 30)$ – ahead of and to the side of the obstacle – and sweep the relative velocity v from -10 to 10 kmph along several directions $(\Delta v_x, \Delta v_y)$, with v scaling along each direction. Negative v corresponds to the vehicle receding, positive v to it closing.

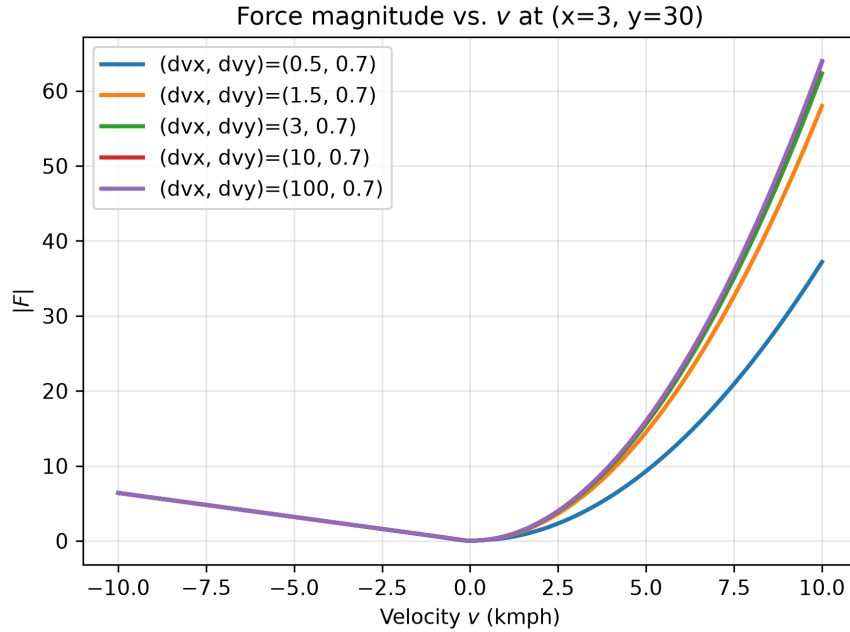


Figure 2: Force magnitude vs. relative velocity at $(x, y) = (3, 30)$.

3 Removing the $\Delta v / \Delta v_{max}$ Term

From Figure 2, we observe that when the vehicle is receding ($v < 0$), an increase in the magnitude of the relative velocity causes the force to increase as well. This is an undesirable effect, since a receding vehicle should not produce a growing repulsive force.

To avoid this behaviour, we drop the $\Delta v / \Delta v_{max}$ term from the original potential function in Equation 1. Repeating the same experiment with this modified potential gives the plot below.

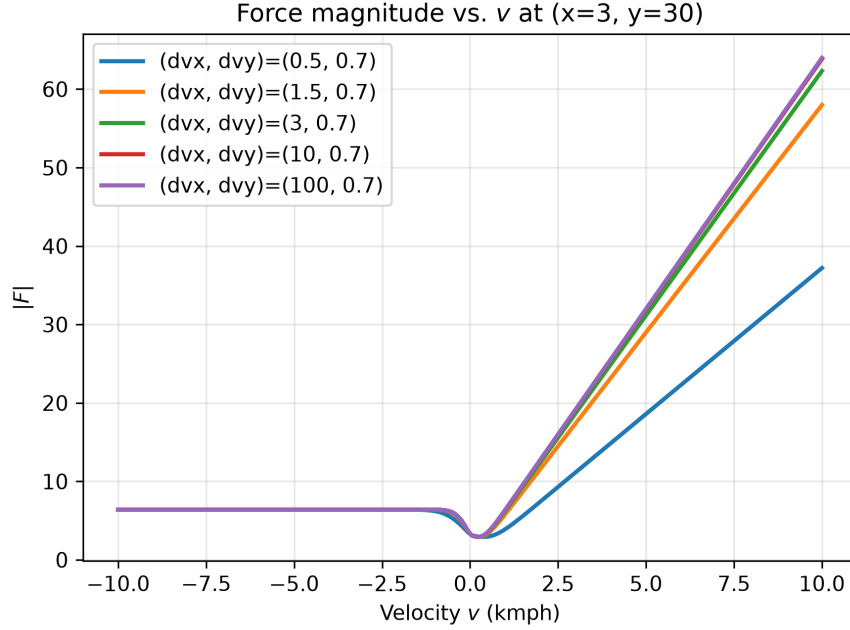


Figure 3: Force magnitude vs. relative velocity at $(x, y) = (3, 30)$, with the $\Delta v / \Delta v_{max}$ term removed.

4 An Alternative Pseudo-Distance Scaling

The time constants $\tau(\Delta x, \Delta v_x)$ and $\tau(\Delta y, \Delta v_y)$ in Equation 1 modulate the x - and y -scales independently, each with its own relative velocity. As an alternative, we keep τ_x, τ_y constant and instead scale the *combined* pseudo-distance by a single factor that depends on the total relative speed and the closing rate:

$$U''_{obst} = \frac{\lambda g(\Delta x, \Delta y, \Delta v_x, \Delta v_y)}{\left(1 - \frac{w}{\Delta x}\right) \sqrt{\left(\frac{\Delta x}{\tau_x}\right)^2 + \left(\frac{\Delta y}{\tau_y}\right)^2}} \quad (4)$$

$$g(\Delta x, \Delta y, \Delta v_x, \Delta v_y) = \frac{(1 + \alpha|\Delta v|) + (\alpha|\Delta v| - 1) \tanh(-(\Delta v_x \Delta x + \Delta v_y \Delta y))}{2}, \quad (5)$$

$$|\Delta v| = \sqrt{\Delta v_x^2 + \Delta v_y^2} \quad (6)$$

As in Section 3, the $\Delta v / \Delta v_{max}$ prefactor is dropped. The resulting force field, under the same three velocity scenarios as Figure 1, is shown below.

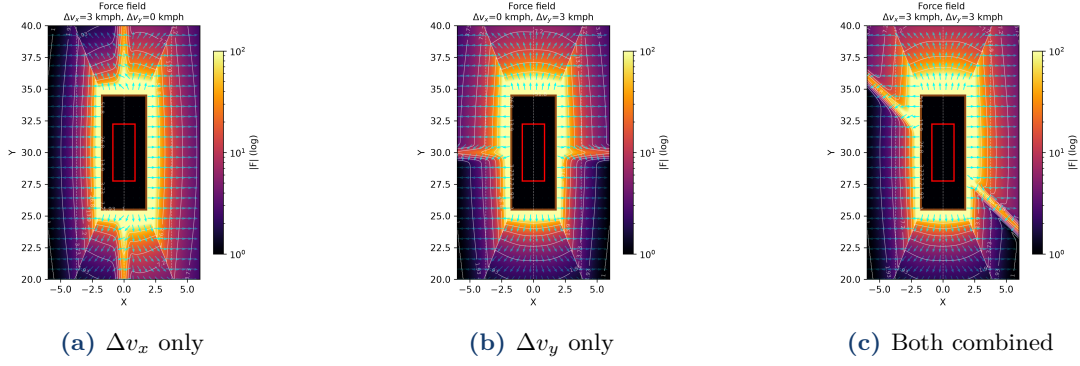


Figure 4: Force field of the potential in Equation 4.

Repeating the fixed-point sweep of Section 2 at $(x, y) = (3, 30)$ with this potential gives:

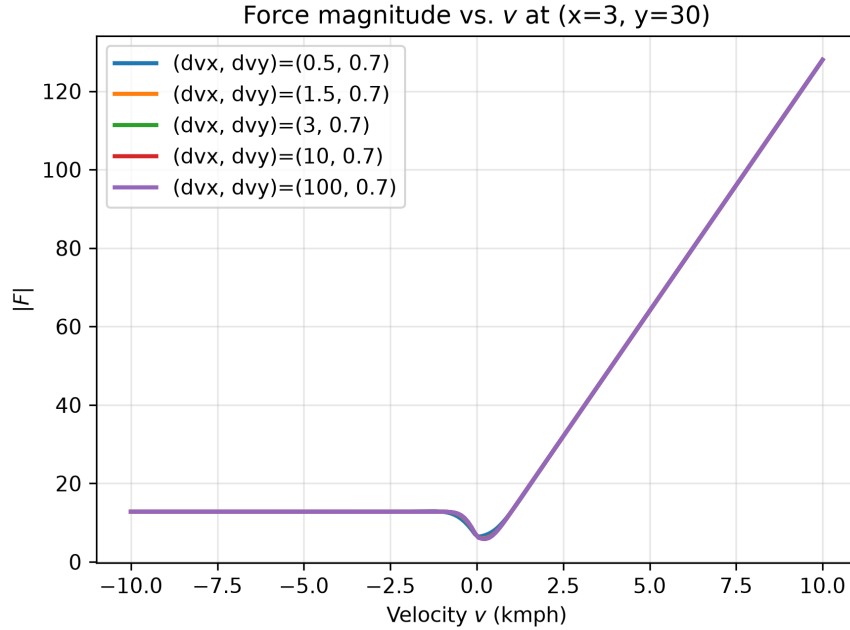


Figure 5: Force magnitude vs. relative velocity at $(x, y) = (3, 30)$, under the potential in Equation 4.

Because the scaling factor g depends only on the combined speed $|\Delta v|$ and the closing rate $\Delta v_x \Delta x + \Delta v_y \Delta y$, rather than on Δv_x and Δv_y separately, the five direction curves of Figure 3 collapse onto essentially one curve in Figure 5. We also note that the plateau at large negative v persists here as well, which suggests that the receding-force effect discussed in Section 3 originates in the tanh modulation of the pseudo-distance itself, rather than in the $\Delta v / \Delta v_{max}$ prefactor alone.